

Trace-Driven Evaluation of Post-Quantum Authentication for ADS-B Using OpenSky Network Data

Marcos Cabañas Maldonado ^{*,1} Junzi Sun ^{1,2} and Martin Strohmeier ^{3,2}

¹Faculty of Aerospace Engineering, Delft University of Technology, Delft, Netherlands

²OpenSky Network, Burgdorf, Switzerland

³Cyber-Defence Campus, armasuisse Science & Technology, Zurich, Switzerland

*Corresponding author: marcoscabanas@outlook.com

1. Background

The emergence of quantum computing poses a long-term threat to the public-key cryptographic algorithms that provide authentication and trust in digital communication systems. This challenge is significant for aviation, where aircraft and air traffic management (ATM) infrastructure are expected to remain operational for several decades, significantly exceeding the transition period to quantum-resistant cryptography. While the recent standardization of certain post-quantum cryptographic (PQC) algorithms provides viable theoretical replacements for classical public-key primitives, their practical deployment in aviation remains largely unexplored. Unlike conventional IT systems, aviation communication and surveillance systems operate under strict operational constraints, including limited bandwidth, fixed protocol structures, strict latency requirements, long certification cycles, and global interoperability. Consequently, the suitability of PQC algorithms cannot be assessed solely on cryptographic security or computational performance, but must instead be evaluated within the operational context of the systems in which they are intended to operate.

2. Methodology

This work builds upon a broader constraint-based assessment of PQC integration across representative aviation communication and surveillance systems, including air-ground datalink, broadcast surveillance, and air-air communication infrastructures. That assessment systematically characterized the operational and technical constraints governing PQC deployment across eight carefully selected systems. Among these, Automatic Dependent Surveillance–Broadcast (ADS-B) emerged as one of the most constrained environments due to its fixed 112-bit message format, high transmission rate, shared 1090 MHz broadcast channel, and absence of native authentication mechanisms [1]. These characteristics make ADS-B a representative case study for evaluating the operational feasibility of post-quantum authentication.

Building on these findings, this paper presents a reproducible methodology to quantitatively evaluate the operational impact of standardized PQC digital signatures under realistic ADS-B traffic conditions. The methodology is demonstrated through a comparative evaluation of three representative candidate signature schemes identified during the broader system assessment: the lattice-based ML-DSA and FN-DSA algorithms [2], and the hash-based SLH-DSA algorithm [3]. Their operational performance is compared against the currently deployed ECDSA signature scheme to establish a practical baseline for evaluating the implications of post-quantum migration. Rather than proposing a new authentication protocol, the objective is to determine which candidate algorithms are compatible with the operational constraints imposed by ADS-B and, for those that are, assess their practical suitability under representative traffic and computational conditions.

The proposed methodology follows a two-stage evaluation process. First, candidate algorithms are screened against the fundamental communication constraints of ADS-B to identify whether they are theoretically compatible with the system. Algorithms that satisfy these constraints are subsequently evaluated through a replay-based simulation using historical ADS-B observations from the OpenSky Network [4]. The overall experimental methodology is summarized in Table 1.

Table 1. Design of the trace-driven evaluation methodology.

Evaluation Stage	Parameter	Description
Stage I – Constraint Screening	Candidate algorithms	ECDSA (baseline), ML-DSA, FN-DSA, and SLH-DSA.
	Screening criteria	Protocol compatibility, signature size, authentication fragmentation requirements, and communication overhead.
	Objective	Identify algorithms theoretically compatible with ADS-B prior to operational evaluation.
Stage II – Operational Evaluation	Traffic environment	Historical OpenSky ADS-B traces under representative traffic densities.
	Authentication architecture	Immediate (sign-before-transmit) or detached (transmit first, authenticate afterwards).
	Authentication interval	Authentication every 1, 5, 10, or 20 ADS-B messages.
	Computational model	Representative airborne signing and ground-receiver verification benchmark profiles.
	Evaluation metrics	Authentication latency, transmission overhead, channel occupancy, verification throughput, and authentication success.
	Objective	Determine the practical suitability of each candidate algorithm under representative operational conditions.

Historical raw ADS-B messages obtained from the OpenSky Network serve as the baseline traffic trace, preserving the temporal characteristics of real aircraft transmissions. Each candidate algorithm identified during the constraint-screening stage is integrated into the trace-driven simulation, where observed ADS-B messages are digitally signed using synthetic cryptographic identities assigned to individual aircraft solely for simulation purposes. Representative signing and verification delays are derived from published embedded-system benchmark data and injected into the replayed traffic rather than measured on the host computer executing the experiment. Authentication information is subsequently incorporated into the replayed traffic according to configurable authentication architectures and authentication intervals, allowing the additional computational and communication burden introduced by each candidate algorithm to be evaluated while preserving the original traffic timeline.

The modified traffic is analyzed through a replay-based discrete-event simulation that models the transmission, reception, and verification of authentication information alongside the original ADS-B broadcasts. This enables the quantitative evaluation of end-to-end authentication latency, signature transmission overhead, additional channel occupancy, verification throughput, signature reconstruction success, and authentication workload across varying traffic densities and authentication strategies. By combining operational surveillance data with standardized cryptographic algorithms and representative computational models, the proposed methodology provides a reproducible, system-level framework for assessing the practical feasibility of post-quantum authentication in ADS-B while remaining independent of any specific implementation platform.

3. Expected results

A central objective of this work is reproducibility. The complete experimental workflow is based on openly available surveillance data, standardized PQC algorithms, published benchmark datasets, and open-source software. The replay simulator, processing scripts, experiment configuration, and generated datasets will be released to facilitate independent reproduction, validation, and future comparative studies involving alternative post-quantum signature schemes and authentication strategies.

Beyond the comparative evaluation of ML-DSA, FN-DSA, and SLH-DSA for ADS-B, the proposed methodology establishes a reproducible framework for assessing post-quantum authentication within operational aviation environments. While this paper focuses on the most constrained surveillance case identified during the broader system assessment, the methodology is sufficiently generic to support future evaluations of additional PQC algorithms and other aviation communication and surveillance systems where representative operational traffic data are available.

References

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